

QUESTION 1

Marks

- (a) If $z = 4 + 3i$ and $w = 2 - i$. Find in simplest exact form:

3

(i) $\frac{z}{w}$

(ii) $\text{Im}(wz)$

(iii) $|3z - 3iw|$

- (b) Indicate on an Argand diagram the region which contains the point P representing z when:

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(i) $|z| > 2$ and $\arg z \leq \frac{\pi}{2}$.

(ii) $|z - 3 - i| > 2$ and $\text{Re}(z) \geq 3 \text{Im}(z)$.

- (c) w is a complex cube root of unity.

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- (i) Show that w^2 is also a root and that $1 + w + w^2 = 0$.

- (ii) Prove that $(1 + w)(1 + 2w)(1 + 3w)(1 + 5w) = 21$.

- (iii) If w and w^2 are roots of $P(x) = x^3 + px^2 + qx + w = 0$, deduce that $p = q = w + 1$.

- (iv) If r is a cube root of a , where $a \in \mathbb{C}$, show that rw and rw^2 are the two cube roots of a .

- (v) Hence, determine the complex numbers z such that

$$\left(\frac{z+1}{z}\right)^3 + 8 = 0.$$

QUESTION 2

Begin a new page.

Marks

- (a) α, β and δ are the roots of $x^3 + px + q = 0$.

3

Evaluate $(\alpha - \beta)^2 + (\beta - \delta)^2 + (\delta - \alpha)^2$.

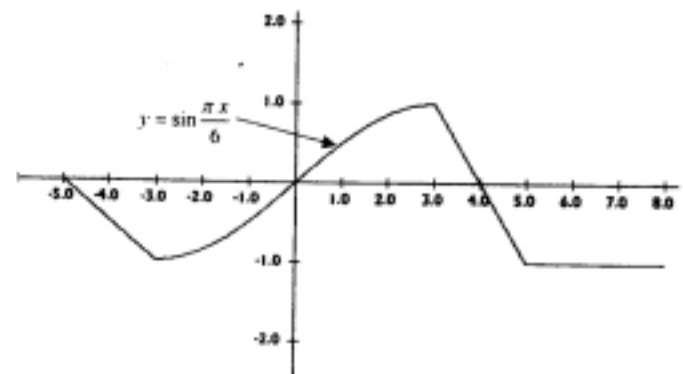
- (b) (i) Prove that if a polynomial $P(x)$ has a root of multiplicity m , then $P'(x)$ has a root of multiplicity $(m - 1)$.

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- (ii) Find the values of k so that the equation $5x^3 - 3x^2 + k = 0$ has two equal roots, both positive.

- (c) The diagram is a sketch of $y = f(x)$ which includes part of the curve $y = \sin \frac{\pi x}{6}$.

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On separate diagrams, sketch each of the following:

(i) $y = -f(x)$

(ii) $y = |f(x)|$

(iii) $y = [f(x)]^2$

(iv) $|y| = f(x)$

QUESTION 7 Begin a new page.**Marks**

- (a) (i) Sketch $y = e^x - 1 - x$ showing all stationary point(s) and asymptote(s). 4
- (ii) Hence, solve $1 + x < e^x$.
- (b) P and Q are on the same branch of the rectangular hyperbola $xy = c^2$. 11
- (i) Show that the equation of the chord joining the points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ is $x + pqy = c(p + q)$.
- (ii) Deduce the equation of the tangent at P.
- (iii) The tangent at P meets the x and y axis at L and M respectively. O is the origin and POD is a diameter. The line MD meets the x axis at T. Prove that the area of triangle DOT is equal to $\frac{c^2}{3}$ square units.
- (iv) The normal at Q meets the x axis at A and the tangent at Q meets the y axis at B. Find the equation of the locus of the mid-point of AB.

QUESTION 8 Begin a new page.**Marks**

- (a) (i) Find the derivative of $y = \ln \sqrt{\frac{1 - \sin x}{1 + \sin x}}$. 3
- (ii) Hence, find $\int \sec x \, dx$.
- (b) (i) Find $\frac{d}{dx} (\cot^{-1} x)$. 3
- (ii) Prove that the function $f(x) = \cot^{-1} x + \tan^{-1} x$ is constant and find the value of this constant.
- (c) Find the coordinates of the point on the graph $x^2 y + xy^2 = 16$ at which the tangent is parallel to the x axis. 4
- (d) Consider the complex number $z = x + iy$ represented by point A on the Argand diagram. 5
- Let $Z = \frac{z - \bar{z} + 2}{z + 1 - i}$ be represented by point M.
- (i) Find the locus of A for which Z is a real number.
- (ii) Find the locus of M as A moves on the x axis.



2000 4 UNIT TRIAL - FSHS - Solution

Question 1.

(a) $z = 4 + 3i$ $w = 2 - i$ $\bar{w} = 2 + i$

(i) $\frac{z}{w} = \frac{4+3i}{2-i} \cdot \frac{(2+i)}{(2+i)} = \frac{8-4i+6i+3}{4+1} = \frac{11}{5} + \frac{2i}{5}$

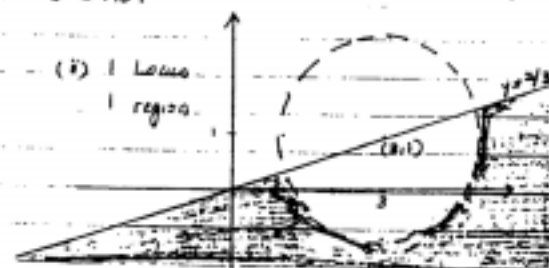
(ii) $wz = (3+4i)(2-i) = 11 + 2i$ $\text{Im}(wz) = 2$

(iii) $|3z - 3i\bar{w}| = |3(4+3i) - 3i(2-i)| = |12+9i-6i-3i^2|$
 $= |9+3i| = \sqrt{9^2+3^2} = \sqrt{90} = 3\sqrt{10}$

(b) (i)



$|z| > 2$ and $\arg z \leq \pi/2$



$|z-3-i| > 2$ is region outside circle of centre $(3,1)$, and $r=2$.

$\text{Re}(z) \geq 3 \text{Im}(z) \Rightarrow x \geq 3y \Rightarrow y \leq \frac{x}{3}$

(c) $w^3 = 1$, w is complex ($w \neq 1$)

$(w^3)^3 = (w^3)^2 = (1)^3 = 1$

$\therefore w^3$ is also a root of $w^3 = 1$.

$(w^3-1) = 0 \Rightarrow (w-1)(w^2+w+1) = 0$

Since $w \neq 1 \Rightarrow w^2+w+1 = 0$ * (See 3)

(ii) $(1+w)(1+2w) - (1+3w) - (1+5w)$

$= (1+3w+2w^2) - (1+8w+15w^2)$

$= (1+2w+2w^2+w) - (1+8w+8w^2+7w^2)$ But $w^2+w = -1$

$= (1+2(-1)+w) - (1+8(-1)+7w^2)$

$= (w-1) - (7w^2-7) = 7(w-1)(w^2-1)$ Since $w^3 = 1$

$= 7(w^2-w^3-w+1) = 7(1-(w^3+w)+1)$ [OR: $(w-1)(7w-17)$ expansion]
 $= 7(1+1-w) = 7 \times 3 = 21$

Full expansion: $1+9w^2+11w + (15+16)w^2+30w = 1+41w+41w^2+61$
 $= 62+41(w+w^2) = 62-41 = 21$

2000 FSHS 4 Unit Trial Solutions

* Alternative sol for (i): $z^3 = 1$ Cube Roots of unity: $\text{cis } 3\theta = \text{cis } 0$

$3\theta = 2k\pi \Rightarrow \theta = 2k\pi/3$ $k = 0, \pm 1$

$\therefore$ cube roots of unity are: 1, $\text{cis } 2\pi/3$, $\text{cis } 4\pi/3 = \text{cis } (-2\pi/3)$

$z^3 - 1 = 0 \Rightarrow (z-1)(z^2+z+1) = 0$

$\therefore$ complex roots must satisfy $z^2+z+1 = 0$

$w = \text{cis } 2\pi/3 \Rightarrow w^2 = \text{cis } 2 \times 2\pi/3 = \text{cis } 4\pi/3$ (By De Moivre's)

But $\text{cis } 4\pi/3 = \text{cis } (-2\pi/3)$

or: $w = \text{cis } 2\pi/3 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$
 $w^2 = \text{cis } (-2\pi/3) = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$ $\Rightarrow 1+w+w^2 = 0$

iii) $P(w) = w^3 + pw^2 + qw + m = 0 \Rightarrow 1 + pw^2 + qw + m = 0$

$P(w^2) = (w^2)^3 + p(w^2)^2 + q(w^2) + m = 0$ Since $(w^2)^3 = (w^3)^2 = 1$
 $= 1 + pw + qw^2 + m = 0$

$P(w) - P(w^2) = 0 \Rightarrow p(w^2-w) + q(w-w^2) = 0$

$p(w^2-w) - q(w^2-w) = 0$

$(p-q)(w^2-w) = 0 \Rightarrow p=q$ since $w^2 \neq w$

But $1 + pw^2 + qw + m = 0 \Rightarrow 1 + pw^2 + pw + m = 0$

$1 + p(w^2+w) + m = 0$

$1 + p(-1) + m = 0 \Rightarrow p = m+1$

$\therefore p = q = m+1$

or: since $-p = w + w^2 = -1-w$ $\therefore p = w+1$ $\text{Exp: } q = w^2 + w + w = 1 + w(w+w^2) = 1-w$
 product $w^2w = w^3 = 1 \Rightarrow w = -m \Rightarrow p = q = 1-w = 1+m$

(iv) $r^3 = a$

$(rw)^3 = r^3w^3 = r^3(1) = a$ since $w^3 = 1$ $\therefore (rw)$ is a cube root

$(r^2w)^3 = r^3w^6 = r^3(w^3)^2 = r^3(1)^2 = r^3 = a \Rightarrow (r^2w)$ is also a cube root
 or: if r is a cube root, next root is $\text{cis } 2\pi/3 r$ or r ; other root is $\text{cis } 4\pi/3 r = w^2 r$

(v) $r^3 = -8$ Roots are -2 ; $-2w = -2\text{cis } 2\pi/3 = -2(-\frac{1}{2} + i\frac{\sqrt{3}}{2})$; $-2w^2 = -2(-\frac{1}{2} - i\frac{\sqrt{3}}{2})$

$\frac{z+1}{2} = -2$

$\frac{z+1}{2} = -2z$

$3z = -1$

$z = -\frac{1}{3}$

$\frac{1}{2}$

$\frac{z+1}{2} = -2(-\frac{1}{2} + i\frac{\sqrt{3}}{2})$

$\frac{z+1}{2} = 1 - \sqrt{3}i$

$z+1 = 2 - \sqrt{3}i$

$z = 1 - \sqrt{3}i$

$z = -\frac{1}{\sqrt{3}i} = \frac{i}{\sqrt{3}}$

$z = -\frac{i}{\sqrt{3}}$

$\frac{z+1}{2} = -2(-\frac{1}{2} - i\frac{\sqrt{3}}{2})$

$\frac{z+1}{2} = 1 + i\sqrt{3}$

$z+1 = 2 + i\sqrt{3}$

$z = \frac{1}{\sqrt{3}} = \frac{i}{\sqrt{3}}$

$z = -\frac{i}{\sqrt{3}}$

$z = -\frac{i}{\sqrt{3}}$

Question 2.

$$(a) x^2 + px + q = 0. \rightarrow \alpha + \beta + \delta = 0.$$

$$\alpha\beta + \alpha\delta + \beta\delta = +p.$$

$$\begin{aligned} \frac{1}{3} I &= (\alpha - \beta)^2 + (\beta - \delta)^2 + (\delta - \alpha)^2 \\ &= (\alpha^2 + \beta^2 - 2\alpha\beta) + (\beta^2 + \delta^2 - 2\beta\delta) + (\delta^2 + \alpha^2 - 2\alpha\delta) \\ &= 2(\alpha^2 + \beta^2 + \delta^2) - 2(\alpha\beta + \beta\delta + \alpha\delta) \\ \text{But } (\alpha + \beta + \delta)^2 &= \alpha^2 + \beta^2 + \delta^2 + 2(\alpha\beta + \beta\delta + \alpha\delta) \\ \Rightarrow \alpha^2 + \beta^2 + \delta^2 &= (\alpha + \beta + \delta)^2 - 2(\alpha\beta + \beta\delta + \alpha\delta) \\ \therefore I &= 2[(\alpha + \beta + \delta)^2 - 2(\alpha\beta + \beta\delta + \alpha\delta)] - 2(\alpha\beta + \beta\delta + \alpha\delta) \\ &= 2[(\alpha + \beta + \delta)^2 - 2(+p)] - 2(+p) \\ &= -4p - 2p \\ &= -6p. \end{aligned}$$

(b) (i) Let a be the n -fold root of $P(x)$.

$$P(x) = (x-a)^n Q(x).$$

$$P'(x) = n(x-a)^{n-1} Q(x) + (x-a)^n Q'(x).$$

$$= (x-a)^{n-1} [nQ(x) + Q'(x)(x-a)].$$

$$= (x-a)^{n-1} [S(x)].$$

Hence " a " is a root of multiplicity $(n-1)$ of $P'(x)$.

$$(ii) P(x) = 5x^5 - 3x^3 + k = 0.$$

$$\text{For } 2 = \text{roots} \Rightarrow P'(x) = 0.$$

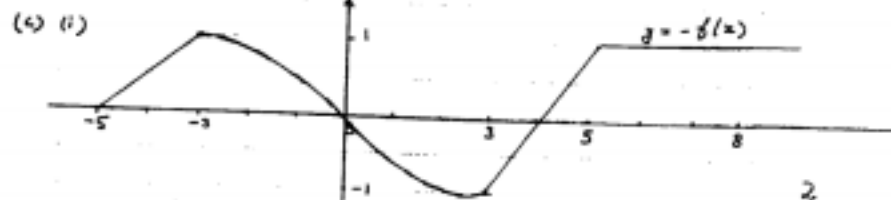
$$P'(x) = 25x^4 - 9x^2 = x^2(25x^2 - 9) = 0.$$

$$x = 0 \text{ or } x = \pm \frac{3}{5}.$$

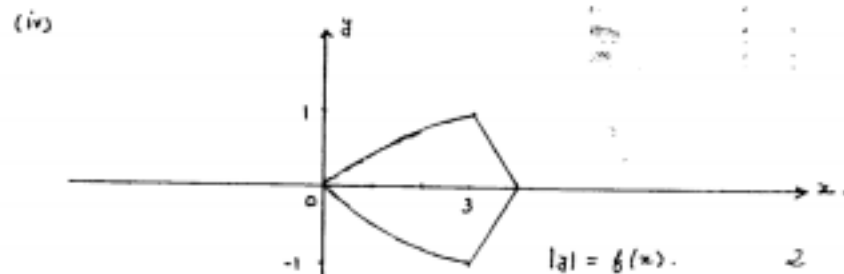
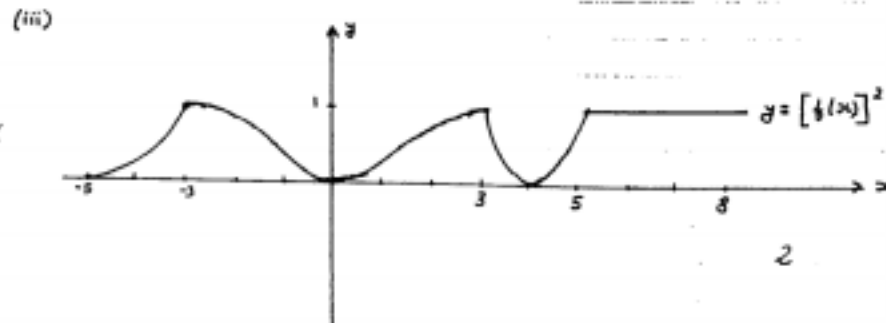
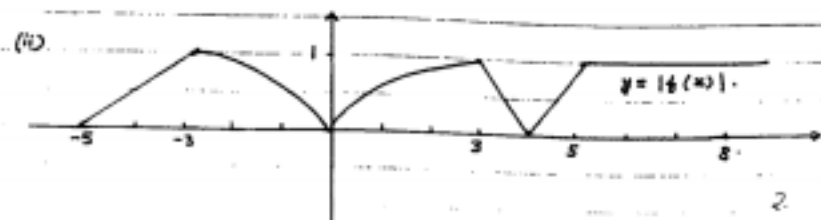
$$\text{But root must be } > 0 \Rightarrow x = \frac{3}{5}.$$

$$\therefore P\left(\frac{3}{5}\right) = 0$$

$$5\left(\frac{3}{5}\right)^5 - 3\left(\frac{3}{5}\right)^3 + k = 0. \Rightarrow k = \frac{3 \times 3^3}{5^3} - 5\left(\frac{3}{5}\right)^5 = \frac{162}{265}.$$



$y = -f(x)$ is a reflection of $y = f(x)$ about the x -axis.

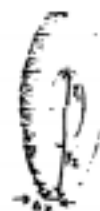
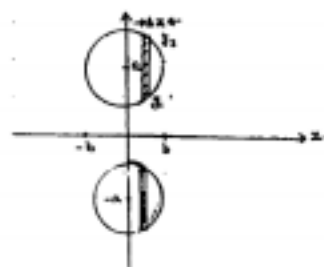


consists of the part of $f(x)$ above the x -axis and its reflection in the x -axis.

2000 F.S.H.S. 4 Unit Trial Solution.

Question 3.

(a) (i) $x^2 + (y-a)^2 = b^2$ where $a > b$.



(ii) $A = \pi (y_2^2 - y_1^2)$.

To find $y_2^2 - y_1^2$????

$(y-a)^2 = b^2 - x^2$.

$y-a = \pm \sqrt{b^2 - x^2}$.

$y = a \pm \sqrt{b^2 - x^2}$.

$\therefore y_2 = a + \sqrt{b^2 - x^2}, y_1 = a - \sqrt{b^2 - x^2}$.

$y_2^2 - y_1^2 = (y_2 + y_1)(y_2 - y_1)$.

$= (a + \sqrt{b^2 - x^2} + a - \sqrt{b^2 - x^2})(a + \sqrt{b^2 - x^2} - a + \sqrt{b^2 - x^2})$.

$= 2a(2\sqrt{b^2 - x^2})$.

$= 4a\sqrt{b^2 - x^2}$.

$\therefore A = \pi[4a\sqrt{b^2 - x^2}] = 4\pi a\sqrt{b^2 - x^2}$.

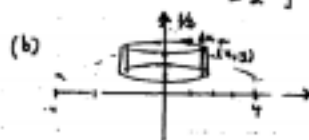
(ii) $\Delta V = 4\pi a\sqrt{b^2 - x^2} \Delta x$ is volume of a slice.

$\therefore V = \sum_{x=-b}^b 4\pi a\sqrt{b^2 - x^2} \Delta x$.

$= \int_{-b}^b 4\pi a\sqrt{b^2 - x^2} dx = 4\pi a \int_{-b}^b \sqrt{b^2 - x^2} dx$.

But $\int_{-b}^b \sqrt{b^2 - x^2} dx$ is area of semi-circle of radius $b = \frac{\pi b^2}{2}$.

$\therefore V = 4\pi a \left[\frac{\pi b^2}{2} \right] = 2\pi^2 ab^2$ (see page 6)



2000 F.S.H.S. 4 Unit Trial Solution.

$R_n = 2\pi xy = 2\pi x \times \frac{1}{x+2} = \frac{2\pi x}{x+2}$

$\Delta V = R_n \Delta x$.

$= \frac{2\pi x}{x+2} \Delta x$ is volume of slice.

$V = \sum_{x=0}^4 \frac{2\pi x}{x+2} \Delta x = \int_0^4 \frac{2\pi x}{x+2} dx = 2\pi \int_0^4 \frac{x}{x+2} dx$.

$= 2\pi \int_0^4 \left(\frac{x+2}{x+2} - \frac{2}{x+2} \right) dx = 2\pi \left[x - 2 \ln|x+2| \right]_0^4$.

$= 2\pi [4 - 2 \ln 6 - 0 + 2 \ln 2] = 2\pi [4 + 2 \ln 2 - 2 \ln 6]$.

$= 2\pi [4 + 2(\ln 2 - \ln 6)] = 2\pi [4 + 2 \ln \frac{2}{6}]$.

$= 8\pi + 4\pi \ln \frac{1}{3} = 4\pi (2 + \ln \frac{1}{3}) = 4\pi (2 + \frac{1}{2} \ln \frac{1}{9})$.

$= 4\pi (2 - \ln 3) \quad \text{or} \quad 2\pi [4 - \ln 9]$.

(c) (i) $I_n = \int_0^1 x^n e^x dx$.

$u = x^n$

$dv = e^x dx$

$du = nx^{n-1}$

$v = e^x$

$I_n = [x^n e^x]_0^1 - n \int_0^1 x^{n-1} e^x dx$.

$I_n = e - 0 - n I_{n-1}$.

$\therefore e = I_n + n I_{n-1} \text{ for } n \geq 1$.

(ii) $I_n = e - n I_{n-1}$.

$I_4 = e - 4 I_3$

$= e - 4(e - 3 I_2) = -3e + 12 I_2$.

$= -3e + 12(e - 2 I_1)$

$= -3e + 12e - 24 I_1$

$= 9e - 24(e - I_0)$

$= -13e + 24 I_0 = -13e + 24(e - 1) = 9e - 24$.

(d) (ii) Long way to find volume:

$\int_{-\pi/2}^{\pi/2} \sqrt{b^2 - x^2} dx$ use $x = b \sin \theta \Rightarrow dx = b \cos \theta d\theta$.

$V = 4\pi a \int_{-\pi/2}^{\pi/2} \sqrt{b^2(1 - \sin^2 \theta)} \cdot b \cos \theta d\theta = 4\pi b^2 a \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta = 4\pi b^2 a \int_{-\pi/2}^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta = 2\pi b^2 a \left[\theta + \frac{1}{2} \sin 2\theta \right]_{-\pi/2}^{\pi/2} = 2\pi b^2 a \left[\frac{\pi}{2} + 0 - \left(-\frac{\pi}{2} + 0 \right) \right] = 2\pi b^2 a \pi = 2\pi^2 ab^2$.

2000 F.S.H.S. 4 Unit Trial Solution.

Question 4.

$$a=3, \quad b=2, \quad \text{so } \frac{x^2}{9} + \frac{y^2}{4} = 1.$$

$$(i) P(3\cos\theta, 2\sin\theta).$$

$$\therefore x_P = 3\cos(\theta + \frac{\pi}{2}) = -3\sin\theta.$$

$$y_P = 2\sin(\theta + \frac{\pi}{2}) = 2\cos\theta.$$

$$\therefore Q(-3\sin\theta, 2\cos\theta).$$

$$(ii) OP^2 + OQ^2 = (3\cos\theta)^2 + (2\sin\theta)^2 + (-3\sin\theta)^2 + (2\cos\theta)^2.$$

$$= 9\cos^2\theta + 4\sin^2\theta + 9\sin^2\theta + 4\cos^2\theta.$$

$$= 9(\cos^2\theta + \sin^2\theta) + 4(\sin^2\theta + \cos^2\theta).$$

$$= 9 \times 1 + 4 \times 1 = 13.$$

$$(iii) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is equation of tangent.}$$

$$\text{at } P: \frac{3\cos\theta \cdot x}{9} + \frac{2\sin\theta \cdot y}{4} = 1. \Rightarrow \frac{\cos\theta \cdot x}{3} + \frac{y \sin\theta}{2} = 1.$$

$$\therefore (2\cos\theta)x + (3\sin\theta)y = 6. \quad \dots (1)$$

$$\text{at } Q: \frac{-3\sin\theta \cdot x}{9} + \frac{(2\cos\theta)y}{4} = 1.$$

$$\frac{-(\sin\theta)x}{3} + \frac{(\cos\theta)y}{2} = 1.$$

$$(-2\sin\theta)x + (3\cos\theta)y = 6. \quad \dots (2)$$

(iv) Solve (1) and (2) simultaneously:

$$(1) \times \sin\theta: 2\cos\theta \sin\theta x + 3\sin^2\theta y = 6\sin\theta.$$

$$(2) \times \cos\theta: -2\sin\theta \cos\theta x + 3\cos^2\theta y = 6\cos\theta.$$

$$\text{Add: } 3(\sin^2\theta + \cos^2\theta)y = 6(\sin\theta + \cos\theta).$$

$$\therefore 3y = 6(\sin\theta + \cos\theta).$$

$$y = 2(\sin\theta + \cos\theta).$$

$$\text{Sub } y \text{ into (1): } 2\cos\theta x + 3\sin\theta (2)(\sin\theta + \cos\theta) = 6.$$

$$2\cos\theta x + 6\sin^2\theta + 6\sin\theta \cos\theta = 6.$$

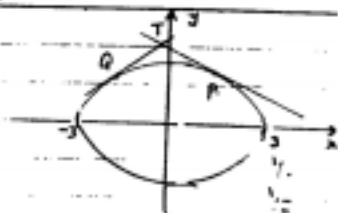
$$2\cos\theta x + 6(1 - \cos^2\theta) + 6\sin\theta \cos\theta = 6.$$

$$2\cos\theta x + 6 - 6\cos^2\theta + 6\sin\theta \cos\theta = 6 \quad \div 2.$$

$$\cos\theta x = 3\cos^2\theta - 3\sin\theta \cos\theta. \quad \div \cos\theta.$$

$$x = 3\cos\theta - 3\sin\theta. \Rightarrow x = 3(\cos\theta - \sin\theta).$$

$$T[3(\cos\theta - \sin\theta), 2(\sin\theta + \cos\theta)].$$



2000 F.S.H.S. 4 Unit Trial Solution.

$$(vi) \frac{x^2}{9} + \frac{y^2}{4} = 1 \Rightarrow \frac{y^2}{4} = 1 - \frac{x^2}{9} \Rightarrow y^2 = 4(1 - \frac{x^2}{9}).$$

$$A = \int_{-3}^3 y \, dx = 2 \int_{-3}^3 \frac{2}{3} \sqrt{9-x^2} \, dx = \frac{4}{3} \int_{-3}^3 \sqrt{9-x^2} \, dx.$$

$$\text{But } \int_{-3}^3 \sqrt{9-x^2} \, dx \text{ is a semi-circle of area } = \frac{\pi \cdot 3^2}{2}.$$

$$\therefore A = \frac{4}{3} \times \pi \times \frac{9}{2} = \frac{36\pi}{3} = 6\pi.$$

(vii)

Area:



$$A = \frac{1}{2} (2x)(2y) \sin 60^\circ.$$

$$= \frac{1}{2} (4x^2) \left(\frac{\sqrt{3}}{2}\right).$$

$$= x^2 \sqrt{3} = 4\sqrt{3} \left(1 - \frac{x^2}{9}\right).$$

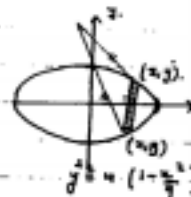
$$dV = x^2 \sqrt{3} \, dx = 4\sqrt{3} \left(1 - \frac{x^2}{9}\right) dx.$$

$$V = \lim_{x \rightarrow 0} \sum_{x=0}^3 4\sqrt{3} \left(1 - \frac{x^2}{9}\right) dx = 4\sqrt{3} \int_0^3 \left(1 - \frac{x^2}{9}\right) dx.$$

$$= 4\sqrt{3} \left[x - \frac{x^3}{27} \right]_0^3 = 4\sqrt{3} \left[3 - \frac{27}{27} - \left(0 - \frac{0}{27}\right) \right].$$

$$= 4\sqrt{3} [3 - 1 + 0 - 0] = 4\sqrt{3} \times 2 = 8\sqrt{3} \text{ unit}^3.$$

$$\text{or } V = 8\sqrt{3} \int_0^3 \left(1 - \frac{x^2}{9}\right) dx = 8\sqrt{3} \left[x - \frac{x^3}{27} \right]_0^3 = 8\sqrt{3} [3 - 1] = 16\sqrt{3} \text{ unit}^3.$$



Question 5.

$$\begin{aligned} \frac{1}{2} \quad (a) \int (4-x) \sqrt{4-x} \, dx &= \int (4-x+5) \sqrt{4-x} \, dx = \int (4-x) \sqrt{4-x} \, dx + 5 \int \sqrt{4-x} \, dx \\ &= \int (4-x)^{3/2} \, dx + 5 \int (4-x)^{1/2} \, dx = \frac{(4-x)^{5/2}}{-5/2} + 5 \frac{(4-x)^{3/2}}{-3/2} + C \\ &= -\frac{2}{5} (4-x)^{5/2} - \frac{10}{3} (4-x)^{3/2} + C. \quad (+ \text{ See page 11}) \end{aligned}$$

$$\frac{1}{3} \quad (b) \int_0^{\pi/2} \frac{dx}{2 \cos x + \sin x} \quad \text{let } t = \tan \frac{x}{2}.$$

$$dx = \frac{1}{1+t^2} \cdot \frac{2}{1+t^2} dt = \frac{2}{(1+t^2)^2} dt$$

$$\Rightarrow dx = \frac{2 dt}{1+t^2}$$

$$I = \int \frac{\frac{2 dt}{1+t^2}}{2 - \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} dt = \int \frac{2 dt}{2+2t^2-1+t^2+2t} = \int \frac{2 dt}{3t^2+2t+1} = \frac{2}{3} \int \frac{dt}{t^2 + \frac{2}{3}t + \frac{1}{3}}$$

$$= \frac{2}{3} \int \frac{dt}{(t + \frac{1}{3})^2 + \frac{2}{9}} = \frac{2}{3} \int \frac{dt}{(t + \frac{1}{3})^2 + (\frac{\sqrt{2}}{3})^2} \quad \frac{1}{2}$$

$$\text{let } u = t + \frac{1}{3} \quad \Rightarrow I = \frac{2}{3} \int \frac{du}{u^2 + \frac{2}{9}} = \frac{2}{3} \cdot \frac{1}{(\frac{\sqrt{2}}{3})} \left[\tan^{-1} \frac{u}{\frac{\sqrt{2}}{3}} \right]_{1/3}^{1/2} \quad \frac{1}{2}$$

$$\text{at } x=0 \Rightarrow u = \tan^{-1} 0 + \frac{1}{3} = \frac{1}{3}$$

$$x = \frac{\pi}{2} \Rightarrow u = \tan^{-1} 1 + \frac{1}{3} = 1 + \frac{1}{3} = \frac{4}{3}$$

$$\therefore I = \sqrt{2} \tan^{-1} \left[\frac{3 \times \frac{4}{3}}{\sqrt{2}} \right] - \sqrt{2} \tan^{-1} \left[\frac{3 \times \frac{1}{3}}{\sqrt{2}} \right] = \sqrt{2} \tan^{-1} \left(\frac{4}{\sqrt{2}} \right) - \sqrt{2} \tan^{-1} \left(\frac{1}{\sqrt{2}} \right)$$

$$= \sqrt{2} \left[\tan^{-1} 2\sqrt{2} - \tan^{-1} \frac{1}{\sqrt{2}} \right]$$

$$\frac{1}{4} \quad (c) A = \int \frac{16x}{x^3-16} dx = \int \frac{16x}{(x^3-4)(x^3+4)} dx = \int \frac{16x}{(x-2)(x+2)(x^2+4)}$$

$$\frac{16x}{(x-2)(x+2)(x^2+4)} = \frac{a}{(x-2)} + \frac{b}{(x+2)} + \frac{cx+d}{x^2+4}$$

$$16x = a(x+2)(x^2+4) + b(x-2)(x^2+4) + (cx+d)(x^2-4)$$

$$\text{let } x=2: 16(2) = a(4)(8) + 0 \Rightarrow 32 = 32a \Rightarrow a=1$$

$$x=-2: -32 = b(-4)(8) + 0 \Rightarrow -32 = -32b \Rightarrow b=1$$

$$\therefore 16x = (x+2)(x^2+4) + (x-2)(x^2+4) + (cx+d)(x^2-4)$$

$$\text{let } x=0: 0 = 2(4) + (-2)(4) + d(-4) \Rightarrow d=0$$

$$x=1: 16 = 3(5) + (-1)(5) + c(-3) \Rightarrow 16-10=6=-3c \Rightarrow c=-2$$

$$\therefore \frac{16x}{(x^3-4)(x^3+4)} = \frac{1}{(x-2)} + \frac{1}{(x+2)} - \frac{2x}{x^2+4} \quad 2 \text{ + 1 = 3}$$

$$A = \int \left[\frac{1}{(x-2)} + \frac{1}{(x+2)} \right] dx - \int \frac{2x}{(x^2+4)} dx$$

$$= \left[\ln|x-2| + \ln|x+2| \right]_4^{10} - \left[\ln x^2 \right]_{20}^{40}$$

$$= \ln 4 + \ln 8 - \ln 2 - \ln 6 - \ln 40 + \ln 20$$

$$= \ln 2^2 + \ln 2^3 - \ln 2 - \ln 6 - [\ln 40 - \ln 20]$$

$$= 2\ln 2 + 3\ln 2 - \ln 2 - \ln 6 - \ln \frac{40}{20}$$

$$= 4\ln 2 - \ln 6 - \ln 2$$

$$= 3\ln 2 - \ln 6 = \ln 2^3 - \ln 6 = \ln 8 - \ln 6$$

$$= \ln \frac{8}{6} = \ln \frac{4}{3}$$

$$\text{OR: } \frac{16x}{(x^3-4)(x^3+4)} = \frac{-2x}{x^2+4} + \frac{2x}{x^2+4}$$

$$\therefore A = \int \left(\frac{-2x}{x^2+4} + \frac{2x}{x^2+4} \right) dx = [\ln(x^2-4) - \ln(x^2+4)]_4^{10} = \left[\ln \frac{x^2-4}{x^2+4} \right]_4^{10}$$

$$= \ln \frac{36}{40} - \ln \frac{12}{20} = \ln \frac{36}{40} \times \frac{20}{12} = \ln \frac{4}{3}$$

$$(d) I = \int \sin^{-1} \frac{x}{a} dx \quad dv = dx \quad u = \sin^{-1} \frac{x}{a}$$

$$v = x \quad du = \frac{1}{a} \cdot \frac{1}{\sqrt{1-\frac{x^2}{a^2}}} dx = \frac{dx}{\sqrt{a^2-x^2}}$$

$$I = uv - \int v du = x \sin^{-1} \frac{x}{a} - \int \frac{x dx}{\sqrt{a^2-x^2}} \quad \text{let } u = a^2-x^2$$

$$= x \sin^{-1} \frac{x}{a} + \int \frac{du}{2\sqrt{u}} = x \sin^{-1} \frac{x}{a} + \frac{1}{2} \cdot \frac{u^{1/2}}{1/2} + C$$

$$= x \sin^{-1} \frac{x}{a} + \sqrt{a^2-x^2} + C$$

$$\text{OR: } u^3 = a^3 - x^2$$

$$2u du = -2x dx$$

$$-u du = x dx$$

$$\int \frac{x dx}{\sqrt{a^3-x^2}} = \int \frac{-u du}{\sqrt{u}} = \int -u^{1/2} du = -\frac{2}{3} u^{3/2} = -\frac{2}{3} (a^3-x^2)^{3/2}$$

$$\therefore I = x \sin^{-1} \frac{x}{a} + u + C$$

$$= x \sin^{-1} \frac{x}{a} + \sqrt{a^3-x^2} + C$$

$$\text{OR: } x = a \sin \theta$$

$$dx = a \cos \theta d\theta$$

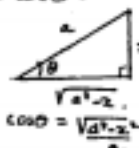
$$= \int \frac{x}{\sqrt{a^3-x^2}} dx = \int \frac{a \sin \theta \cdot a \cos \theta d\theta}{a \cos \theta}$$

$$= \int a \sin \theta d\theta$$

$$= -a \cos \theta$$

$$= -a \left[\frac{\sqrt{a^2-x^2}}{a} \right]$$

$$= -\sqrt{a^2-x^2}$$



2000 FSMS 4 Unit Trial Solutions.

$$(c)(i) \text{ R.T.P. } \int_0^a f(x) dx = \int_0^a f(a-x) dx.$$

$$\text{4 let } u = a-x. \quad x=0 \Rightarrow u=a. \quad x=a \Rightarrow u=0. \quad du = -dx.$$

$$\int_0^a f(x) dx = - \int_a^0 f(a-u) du = \int_0^a f(a-u) du = \int_0^a f(a-x) dx. \quad 2$$

$$(ii) \text{ Let } \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} \frac{\sqrt{\sin(\pi/2-x)}}{\sqrt{\sin(\pi/2-x)} + \sqrt{\cos(\pi/2-x)}} dx = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx.$$

$$\therefore \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx. \quad \text{Add I to both sides.}$$

$$\Rightarrow \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx + \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}.$$

$$2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}.$$

$$2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}.$$

$$\therefore 2I = \left[x \right]_0^{\pi/2} = \frac{\pi}{2}.$$

$$\Rightarrow I = \frac{\pi}{4}.$$

(a) Alternative Slas: (longer ways!)

$$u^2 = 4-x.$$

$$x = 4-u^2.$$

$$dx = -2u du.$$

$$\int (4-u^2) u \cdot (-2u du)$$

$$= -2 \int (4u^2 - u^4) du.$$

$$= -2 \left[\frac{4u^3}{3} - \frac{u^5}{5} \right].$$

$$= -\frac{10}{3} u^3 + \frac{2}{5} u^5 + C.$$

$$= -\frac{10}{3} \sqrt{4-x}^3 + \frac{2}{5} \sqrt{4-x}^5 + C.$$

$$(4-x) \sqrt{4-x} = 4\sqrt{4-x} - x\sqrt{4-x}.$$

$$\int 4\sqrt{4-x} dx - \int x\sqrt{4-x} dx. \quad \text{let } u = 4-x.$$

$$= \frac{4}{3/2} (4-x)^{3/2} + \int (4-u) \sqrt{u} du.$$

$$= -\frac{4}{3/2} (4-x)^{3/2} + \int (4-u) \sqrt{u} du.$$

$$= -\frac{4}{3/2} (4-x)^{3/2} + \frac{4}{3/2} (4-x)^{3/2} - \frac{2}{5/2} (4-x)^{5/2}.$$

$$= -\frac{10}{3} (4-x)^{3/2} + \frac{2}{5} (4-x)^{5/2} + C.$$

2000 FSMS 4 Unit Trial Solutions.

Question 6.

$$(a)(i) (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

Step 1: Prove true for $n=1$.

$$(\cos \theta + i \sin \theta)^1 = \cos \theta + i \sin \theta \Rightarrow \text{true for } n=1. \quad 1/2$$

Step 2: Assume true for $n=k$: $(\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$.

Prove true for $n=k+1$ i.e. $(\cos \theta + i \sin \theta)^{k+1} = \cos (k+1)\theta + i \sin (k+1)\theta$.

$$\text{Proof: } (\cos \theta + i \sin \theta)^{k+1} = (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta)$$

$$= (\cos k\theta + i \sin k\theta) (\cos \theta + i \sin \theta) \quad \text{from assumption.}$$

$$= \cos k\theta \cos \theta - \sin k\theta \sin \theta + i (\cos k\theta \sin \theta + \sin k\theta \cos \theta).$$

$$= \cos (k\theta + \theta) + i \sin (k\theta + \theta).$$

$$= \cos (k+1)\theta + i \sin (k+1)\theta. \quad 2$$

Step 3: Since it's true for $n=1$, it is true for $n=2$. Since it's

true for $n=k$, it's true for $n=k+1$. $1/2$

$\therefore$ It is true for all positive integers k .

(ii) let $c = \cos \theta$ and $s = \sin \theta$.

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta.$$

$$\text{3 } (c+is)^5 = c^5 + 5c^4is + 10c^3(is)^2 + 10c^2(is)^3 + 5c(is)^4 + (is)^5$$

$$= c^5 + 5c^4is - 10c^3s^2 - 10c^2s^3i + 5cs^4 + is^5.$$

$$= (c^5 - 10c^3s^2 + 5cs^4) + i(5c^4s - 10c^2s^3 + s^5) = \cos 5\theta + i \sin 5\theta.$$

$$\therefore \cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta.$$

$$= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2$$

$$= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta (1 + \cos^2 \theta - 2 \cos^2 \theta)$$

$$= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta + 5 \cos^3 \theta - 10 \cos^3 \theta.$$

$$= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta. \quad 2$$

(iii) $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$. Let $x = \cos \theta$.

$$= 16x^5 - 20x^3 + 5x = \sqrt{3}.$$

$$\text{2 } \text{so gives us } 32x^5 - 40x^3 + 10x - \sqrt{3} = 0.$$

$$\therefore \cos 5\theta = \frac{\sqrt{3}}{2} = \cos \frac{\pi}{6}.$$

$$\therefore 5\theta = \frac{\pi}{6} + 2k\pi \quad \text{where } k=0, \pm 1, \pm 2.$$

$$\theta = \frac{\pi}{30} + \frac{2k\pi}{5}.$$

$$\theta = \frac{\pi}{30}; \frac{\pi}{30} + \frac{2\pi}{5} = \frac{13\pi}{30}; \frac{\pi}{30} - \frac{2\pi}{5} = -\frac{11\pi}{30}; \frac{\pi}{30} + \frac{4\pi}{5} = \frac{9\pi}{10}$$

$$\frac{\pi}{30} - \frac{4\pi}{5} = -\frac{23\pi}{30} \text{ or } \frac{37\pi}{30}.$$

$$\therefore \text{solutions are: } \cos \frac{\pi}{30}, \cos \frac{13\pi}{30}, \cos \frac{17\pi}{30}, \cos \frac{5\pi}{6} = -\frac{\sqrt{3}}{2}, \cos \frac{11\pi}{30}.$$

P.S: You can also use: $5\theta = 2k\pi \pm \frac{\pi}{6}$ (general formula of cos = value).

(b)(i) For $x \geq 0$ $[H(x)]^2 = y^2 = (1-x)^2(4-x)$.

2g. $y' = -2(1-x)(4-x) - 1(1-x)^2$
 $= -(1-x)[8-2x+1-x] = (x-1)(9-3x) = 3(x-1)(3-x)$.

$\therefore y' = \frac{3(x-1)(3-x)}{\pm 2(1-x)\sqrt{4-x}} = \pm \frac{3(3-x)}{2\sqrt{4-x}} = 0$.

$\therefore x=3$ is sol of $y'=0$. $\Rightarrow y = \pm\sqrt{4-x} = \pm 2$. $\Rightarrow (3,2) + (3,-2)$. Since $x \neq 0$

For $y = (1-x)\sqrt{4-x}$. $\Rightarrow y' = \frac{3(x-1)(3-x)}{2(1-x)\sqrt{4-x}} = -\frac{3(3-x)}{2\sqrt{4-x}}$.

x	3 ⁻	3	3 ⁺
y'	-	0	+

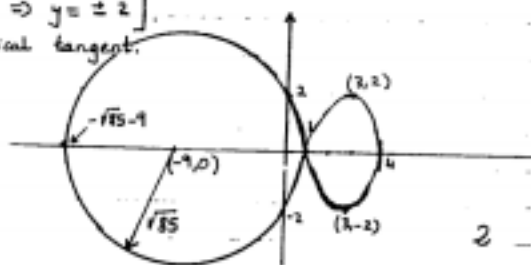
Min. $y = -(1-x)\sqrt{4-x} \Rightarrow y' = \frac{3(3-x)}{2\sqrt{4-x}}$.

x	3 ⁻	3	3 ⁺
y'	+	0	-

(ii) $y^2 = 85 - (x+2)^2$ for $x < 0$ is the equation of a circle
 $(x+2)^2 + y^2 = 85$ of centre $(-2,0)$ and radius $= \sqrt{85}$.

[at $x=0$ $81+y^2=85 \Rightarrow y^2=4 \Rightarrow y=\pm 2$].

For $x=4$ $y'=0 \Rightarrow$ vertical tangent.



(iii) $V = \pi \int_1^4 y^2 dx = \pi \int_1^4 (1-x)^2(4-x) dx$.

2. No $(1-x)^2(4-x) = (x^2-2x+1)(4-x) = (4x^2-8x+4-x^3+2x^2-x) = -x^3+6x^2-9x+4$.

$V = \pi \int_1^4 (-x^3+6x^2-9x+4) dx = \pi \left[-\frac{x^4}{4} + \frac{6x^3}{3} - \frac{9x^2}{2} + 4x \right]_1^4$

$V = \pi \left[-\frac{4^4}{4} + \frac{6 \cdot 4^3}{3} - \frac{9 \cdot 4^2}{2} + 4 \cdot 4 \right] - \left[-\frac{1^4}{4} + \frac{6 \cdot 1^3}{3} - \frac{9 \cdot 1^2}{2} + 4 \cdot 1 \right]$

$V = \pi \left[-64 + 128 - 72 + 16 - \left(-\frac{1}{4} + 2 - \frac{9}{2} + 4 \right) \right] = \pi \left[6\frac{3}{4} \right] = 2\frac{3}{4}\pi$.

Question 7

(a) (i) $y = e^x - 1 - x = e^x - (1+x)$.

$y' = e^x - 1 = 0 \Rightarrow e^x = 1 \Rightarrow x=0$ is a stationary pt. $\Rightarrow y = 1 - 1 = 0$

Nature of (0,0): $y'' = e^x = e^0 = 1 > 0 \Rightarrow$ concave up $\Rightarrow$ minimum (0,0)

Limiting values: as $x \rightarrow +\infty$, $y = e^x [1 - \frac{(1+x)}{e^x}] \rightarrow +\infty$

$x \rightarrow -\infty$, $y = e^{-\infty} - (-\infty) = 0 + \infty \rightarrow +\infty$

The line $y = -x - 1$ is an oblique asymptote as can be shown from addition of the 2 curves: $y = e^x$ and $y = -x - 1$.

More points:

$x=1 \Rightarrow y = e - 2 \approx 0.718$

$x=-1 \Rightarrow y = \frac{1}{e} \approx 0.368$

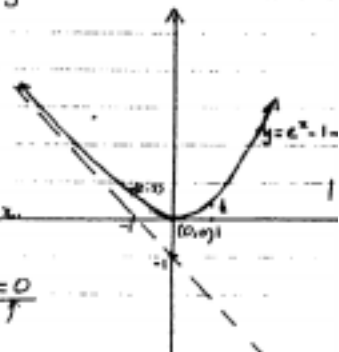
(ii) From the graph we can see:

$e^x - 1 - x \geq 0$ for all real x .

$\Rightarrow e^x \geq 1+x$ or $1+x \leq e^x$ for all real x .

But: $1+x = e^x$ for $x=0$.

$\therefore 1+x < e^x$ for all real x except $x=0$



(b) $xy = c^2 \Rightarrow y = \frac{c^2}{x}$.

(i) $m = \frac{\frac{c}{p} - \frac{c}{q}}{\frac{c}{q} - \frac{c}{p}} = \frac{cp - cq}{pq} \times \frac{1}{cq - cp} = \frac{c(p-q)}{pq(-c)(p-q)} = -\frac{1}{pq}$

$\therefore y - \frac{c}{p} = -\frac{1}{pq}(x - cp) \times pq$

$pqy - cq = -x + cp$

$\Rightarrow x + pqy = cp + cq$

$\therefore x + pqy = c(p+q)$ is the equation of chord PQ.

(ii) P approaches Q and the chord becomes tangent i.e. $p=q$

$\therefore$ tangent at P for $p=q \Rightarrow x + p^2y = c(p+p) = 2cp$.

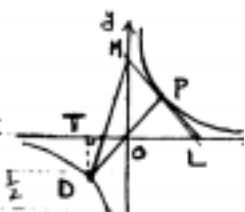
$\therefore$ tangent: $x + p^2y = 2cp$.

(iii) D $(-cp, -\frac{c}{p})$ by symmetry of hyperbola.

4. Tangent at P meets y-axis at M $\therefore x=0$.

$p^2y = 2cp \Rightarrow y = \frac{2c}{p} = \frac{2c}{p} \therefore M(0, \frac{2c}{p})$

gradient of MD: $m = \frac{\frac{2c}{p} - (-\frac{c}{p})}{0 - (-cp)} = \frac{\frac{3c}{p}}{cp} = \frac{3}{p^2} \times \frac{1}{c} = \frac{3}{p^2c}$



2000 FSHS 4 UNIT TRIAL Solutions

Question 8.

$$(a)(i) \quad y = \ln \left(\frac{1 - \sin x}{1 + \sin x} \right)^{\frac{1}{4}} = \frac{1}{4} \ln \left(\frac{1 - \sin x}{1 + \sin x} \right) = \frac{1}{4} [\ln(1 - \sin x) - \ln(1 + \sin x)]$$

$$y' = \frac{dy}{dx} = \frac{1}{4} \left[\frac{-\cos x}{1 - \sin x} - \frac{\cos x}{1 + \sin x} \right] = \frac{1}{4} \left[\frac{-\cos x (1 + \sin x) - \cos x (1 - \sin x)}{1 - \sin^2 x} \right]$$

$$\frac{1}{4} = \frac{1}{4} \left[\frac{-\cos x - \cos x \sin x - \cos x + \cos x \sin x}{\cos^2 x} \right] \quad \text{as } 1 - \sin^2 x = \cos^2 x.$$

$$= -\frac{2\cos x}{4\cos^2 x} = -\frac{1}{2\cos x} = -\frac{1}{2} \sec x$$

$$(ii) \quad \frac{d}{dx} \left[\ln \sqrt{\frac{1 - \sin x}{1 + \sin x}} \right] = -\frac{1}{2} \sec x. \quad (\text{Take primitive of both sides})$$

$$\therefore \int -\frac{1}{2} \sec x \, dx = \ln \sqrt{\frac{1 - \sin x}{1 + \sin x}} + C, \quad x = 2.$$

$$\int \sec x \, dx = -2 \ln \sqrt{\frac{1 - \sin x}{1 + \sin x}} + C.$$

$$\therefore \int \sec x \, dx = -\frac{2}{4} \ln \left(\frac{1 - \sin x}{1 + \sin x} \right) + C = \ln \left(\frac{1 - \sin x}{1 + \sin x} \right)^{-\frac{1}{2}} + C.$$

$$\therefore \int \sec x \, dx = \ln \left(\sqrt{\frac{1 + \sin x}{1 - \sin x}} \right) + C.$$

$$(b)(i) \quad \text{Let } y = \cot^{-1}(x) \Rightarrow x = \cot y.$$

$$\therefore 1 = \cot^2 y \cdot \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{\cot^2 y} = \frac{1}{1 + \cot^2 y} = \frac{1}{1 + x^2}.$$

$$(ii) \quad f(x) = \cot^{-1}(x) + \tan^{-1}(x)$$

$$f'(x) = \frac{-1}{1+x^2} + \frac{1}{1+x^2} = 0.$$

$$\therefore f(x) \text{ is a constant i.e. } f(x) = c \text{ as its derivative} = 0.$$

To find the constant, we know:

$$\tan \frac{\pi}{4} = 1 \Rightarrow \tan^{-1}(1) = \frac{\pi}{4}.$$

$$\cot \frac{\pi}{4} = 1 \Rightarrow \cot^{-1}(1) = \frac{\pi}{4}.$$

$$f(x) = \cot^{-1}(1) + \tan^{-1}(1) = c.$$

$$\frac{\pi}{4} + \frac{\pi}{4} = c \Rightarrow c = \frac{\pi}{2}.$$

$$\therefore f(x) = \tan^{-1}(x) + \cot^{-1}(x) = \frac{\pi}{2}.$$

2000 4 UNIT FSHS Trial Solutions.

$$\text{Equation of MD: } y - \frac{2c}{p} = \frac{2}{p^2} (x - 0) \Rightarrow y - \frac{2c}{p} = \frac{2}{p^2} x.$$

$$\text{MD cuts the } x \text{ axis at } P \therefore y = 0 \therefore -\frac{2c}{p} = \frac{2}{p^2} x.$$

$$2 \text{ for } T \therefore x = -\frac{2cp^2}{3p} = -\frac{2cp}{3} \therefore T \left(-\frac{2cp}{3}, 0 \right).$$

$$\therefore |OT| = \frac{2cp}{3} \text{ units.}$$

$$\text{height of } \Delta PDT = y \text{ coordinate of point } D = \left| -\frac{c}{p} \right| = \frac{c}{p}.$$

$$\therefore \text{Area of } \Delta PDT = \frac{1}{2} \times |OT| \times \text{height} = \frac{1}{2} \times \frac{2cp}{3} \times \frac{c}{p} = \frac{c^2}{3} \text{ units.}$$

$$(iv) \text{ Tangent at } Q: x + y^2 = 2c.$$

$$\text{Meets } y \text{ axis at } B: x = 0 \therefore y = \frac{2c}{1} = 2c \therefore B(0, 2c).$$

$$m_{\text{tangent}} = -\frac{1}{2y}.$$

$$\Rightarrow m_{\text{normal}} = 2y \text{ at } Q(c, \frac{c}{2}).$$

$$\therefore \text{Equation: } y - \frac{c}{2} = 2y^2(x - c) \Rightarrow y^2 x - y = c(y^2 - \frac{1}{2}).$$

$$\text{cuts } x \text{ axis at } A \Rightarrow y = 0 \Rightarrow y^2 x = c(y^2 - \frac{1}{2}).$$

$$\Rightarrow x = \frac{c}{y^2} \left[y^2 - \frac{1}{2} \right] = c - \frac{c}{2y^2} \therefore A \left(c - \frac{c}{2y^2}, 0 \right).$$

Let N be mid-pt. of AB:

$$x_N = \frac{c - \frac{c}{2y^2} + 0}{2} = \frac{c}{2} - \frac{c}{4y^2}.$$

$$y_N = \frac{\frac{2c}{2} + 0}{2} = \frac{c}{2} \Rightarrow y = \frac{c}{y}.$$

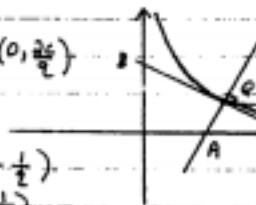
$$\text{Sub } y = \frac{c}{y} \text{ into } x = \frac{c}{y^2} \left[y^2 - \frac{1}{2} \right] = \frac{c}{2} - \frac{c}{4y^2} = \frac{c}{2} - \frac{c}{4 \left(\frac{c^2}{y^2} \right)}.$$

$$\therefore x = \frac{c^2}{2y} - \frac{y^2}{2c^2} \quad \times 2Yc^2.$$

Locus
1/2

$$2YXc^2 = c^4 - Y^4.$$

$$\therefore 2c^2XY + Y^4 = c^4 \text{ is the equation of the locus.}$$



2000 FSHS 4 Unit Trial Solution.

$$(1) \quad x^2y + xy^2 = 16. \quad \dots\dots\dots (1)$$

$$2xy + x^2 \frac{dy}{dx} + y^2 + 2xy \frac{dy}{dx} = 0.$$

$$(x^2 + 2xy) \frac{dy}{dx} = -(2xy + y^2).$$

$$\frac{dy}{dx} = \frac{-(2xy + y^2)}{x^2 + 2xy} = \frac{-y(2x + y)}{x(x + 2y)}$$

$$\text{tangent // to } z \text{ axis} \Rightarrow m = 0 = \frac{dy}{dx}$$

$$\frac{-y(2x + y)}{x(x + 2y)} = 0 \Rightarrow y(2x + y) = 0.$$

$$\therefore y = 0 \text{ or } y = -2x. \dots\dots (2)$$

But $y = 0$ is not on the curve $\therefore y = 0$ is not a tangent.

[Since $y = 0$ is a horizontal asymptote as: $x = \frac{-y^2 \pm \sqrt{y^4 + 64y}}{2y}$ horizontal asymptote]

Sub. (2) into (1).

$$-2x^2 + (2x)^2x = 16 \Rightarrow -2x^2 + 4x^3 = 16 \Rightarrow 2x^3 = 16.$$

$$\Rightarrow x^3 = 8 \Rightarrow x = \sqrt[3]{8} = 2.$$

Sub $x = 2$ into $x^2y + xy^2 = 16$.

$$4y + 2y^2 - 16 = 0 \Rightarrow 2[y^2 + 2y - 8] = 0.$$

$$2[(y-2)(y+4)] = 0.$$

$$\Rightarrow y-2 = 0 \Rightarrow y = 2, \text{ or } y = -4. \dots\dots 2$$

$$\text{at } (2, 2): \frac{dy}{dx} = \frac{-2(4+2)}{2(2+4)} = -1 \neq 0.$$

$$\text{at } (2, -4): \frac{dy}{dx} = \frac{4(4-4)}{2(2-8)} = 0.$$

$\therefore (2, -4)$ is the only pt. where tangent is // to x axis.

2nd method: (Smarter!)

When tangent // to x axis, its equation is $y = a$.

$$\therefore x^2y + xy^2 = 16 \text{ and } y = a \Rightarrow ax^2 + a^2x - 16 = 0.$$

is equation of $\cap$. But for tangency: $\Delta = 0$.

$$\Delta = a^4 + 64a = 0 \Rightarrow a(a^3 + 64) = 0 \Rightarrow a = 0 \text{ or } a^3 = -64.$$

But $a = 0$ rejected as it is x axis. $\Rightarrow a^3 = -64 \Rightarrow a = -4$

$\therefore y = -4$ is the required tangent.

$$-4x^2 + 16x = 16 = 0 \Rightarrow -4(x^2 - 4x + 4) = 0 \Rightarrow -4(x-2)^2 = 0.$$

$$\Rightarrow x = 2$$

2000 4 Unit FSHS Trial Solutions.

$$(d) \quad Z = \frac{z - \bar{z} + 2}{z + 1 - i} = \frac{x + iy - x + i(-y) + 2}{x + iy + 1 - i} = \frac{2iy + 2}{(x+1) + i(y-1)}$$

$$(1) \quad Z = \frac{2(iy+1)}{(x+1) + i(y-1)} \times \frac{(x+1) - i(y-1)}{(x+1) - i(y-1)} = \frac{2y(x+1)i + 2(x+1) + 2y(y-1) - 2i(y-1)}{(x+1)^2 + (y-1)^2}$$

$$= \frac{2x + 2 + 2y^2 - 2y + i[2yx + 2y - 2y + 2]}{(x+1)^2 + (y-1)^2} = \frac{2(x+1+y^2-y) + 2i(xy+1)}{(x+1)^2 + (y-1)^2}$$

$$Z \text{ is real } \Rightarrow \text{Im}(Z) = 0 \Rightarrow \frac{2(x+1+y^2-y)}{(x+1)^2 + (y-1)^2} + \frac{2(xy+1)i}{(x+1)^2 + (y-1)^2}$$

$$\therefore \text{Im}(Z) = 2(xy+1) = 0 \Rightarrow xy = -1 \Rightarrow y = -\frac{1}{x} \text{ But } x \neq 0$$

$\therefore$ Locus of A is a hyperbola $y = -\frac{1}{x}$ except the point $(-1, 1)$.

(ii) A moves on the z axis $\Rightarrow y = 0$ and $z = x = \bar{z}$.

$$\therefore Z = \frac{2(x+1)}{(x+1)^2 + 1} + \frac{2i}{(x+1)^2 + 1} = \frac{2(x+1)}{x^2 + 2x + 2} + \frac{2i}{(x^2 + 2x + 2)}$$

$$\therefore X = \frac{2(x+1)}{(x+1)^2 + 1} \quad Y = \frac{2}{(x+1)^2 + 1}$$

$$\Rightarrow \frac{X}{Y} = \frac{2(x+1)}{2} = x+1. \dots\dots (1)$$

$$\text{But } Y = \frac{2}{(x+1)^2 + 1} \Rightarrow (x+1)^2 + 1 = \frac{2}{Y} \Rightarrow (x+1)^2 = \frac{2}{Y} - 1 \dots\dots (2)$$

$$\text{Square (1): } \frac{X^2}{Y^2} = (x+1)^2 = \left(\frac{2}{Y} - 1\right)$$

$$\therefore \frac{X^2}{Y^2} = \frac{2}{Y} - 1 \quad \times Y^2. \quad \frac{1}{2}$$

$$X^2 = 2Y - Y^2 \Rightarrow X^2 + Y^2 - 2Y + 1 = 1.$$

$$X^2 + (Y-1)^2 = 1.$$

$\therefore$ Locus of M is a circle of radius 1 and centre $(0, 1)$.